LessonPlan Career Roadmap

Science - PCB (Physics, Chemistry, Biology): Class 6 to Career

STATUS: DRAFT FOR HUMAN REVIEW - NOT FOR PUBLIC USE
Drafted 17 Oct 2026 by an AI assistant, then checked the same day by web search (version 2). Sources were mostly coaching-institute and scholarship-aggregator pages plus news reports; official sites (NTA, KEA, DST) could not be opened directly. Items still marked **[VERIFY]** are unconfirmed or conflicting and must be checked on the official site before publishing. Unmarked items are corroborated by at least one secondary source but should still be spot-checked. See section 9 for what changed.

1. Pathway overview

Stage	When	Key decision	What it unlocks
Foundation	Class 6-8	Start Olympiads; build Science and Maths basics	Olympiad ladder, NMMS scholarship (Class 8)
Pre-board	Class 9-10	Maths level (CBSE Basic vs Standard); Junior Science Olympiad	Keeps PCMB open if Standard Maths is taken
Stream choice	After Class 10	Choose PCB (+ optional Maths / Computer Science / Psychology) in Class 11-12 (PUC in Karnataka)	Eligibility for NEET, CUET, IISER, life-science degrees
Preparation	Class 11-12	Boards + entrance exam focus (NEET / KCET / CUET / IAT)	Merit seats, scholarships (INSPIRE-SHE, CSSS)
Degree	After Class 12	Medical, allied health, pharmacy, life sciences, agriculture, research	Professional licence or research career
Postgraduate	3-6 years later	Specialise (MD/MS, MSc, M.Pharm, PhD)	Specialist and research roles
Career	-	Clinical, research, industry, public health, teaching	See section 7

2. Stage-by-stage nodes

Class 6-8: Foundation

- Decision: join Science, Maths and English Olympiads for exposure. Not mandatory, but it builds the profile used later.
- Unlocks: NMMS scholarship exam in Class 8 **[VERIFY state timing]**; INSPIRE-MANAK idea awards (Class 6-10).

Class 9-10: Pre-board

- Decision: in CBSE, choose Standard or Basic Maths for Class 10. PCB itself does not need Standard Maths, but Basic Maths closes the PCMB/Engineering option later. Take Standard if undecided. CBSE relaxed the rule for 2023-24 and 2025-26 (Basic Maths students may take Maths in Class 11 if the school agrees) but it may end with the new scheme. **[VERIFY CBSE circular; Karnataka SSLC rule not found]**
- Junior Science Olympiad (NSEJS) is for Class 10 and below.
- Prerequisite for PCB: Science in Class 10 with good marks in Biology and Chemistry concepts.

After Class 10: Stream choice

- Choose PCB. Add one optional subject: Maths (PCMB, keeps engineering and biotechnology open), Computer Science (bioinformatics), Psychology, or Physical Education.
- Karnataka note: Class 11-12 is the 2-year Pre-University Course (PUC) after SSLC; CBSE/ICSE schools run Class 11-12 directly.
- NEET requires Physics, Chemistry, Biology (or Biotechnology) and English in Class 12.

Class 11-12: Preparation

- Board marks matter: minimum aggregate in PCB is part of NEET eligibility, and KCET ranking uses PUC marks.
- Apply in parallel: NEET-UG, KCET (Karnataka), CUET-UG (central universities), IISER Aptitude Test, NEST.

After Class 12: Degree

- Medical and related: MBBS, BDS, BAMS, BHMS, BUMS, BVSc & AH (all through NEET).
- Allied: Nursing, Pharmacy, Physiotherapy, Medical Lab Technology, Optometry, Radiology.
- Science and research: BSc Life Sciences, Biotechnology, Microbiology, Genetics, Biochemistry, IISER BS-MS.
- Agriculture and environment: BSc Agriculture, Horticulture, Forestry, Fisheries.

3. Olympiads and early exams (Class 6-10)

Exam	Conducted by	Classes	Levels	Tentative timing
National Science Olympiad (NSO)	Science Olympiad Foundation (SOF)	1-12	Level 1 (school) then Level 2 (top performers) **[VERIFY]**	Level 1 Oct-Dec, Level 2 Jan-Mar **[VERIFY]**
International Mathematics Olympiad (IMO)	SOF	1-12	Level 1, Level 2	Similar to NSO **[VERIFY]**
NSEJS then INJSO (Junior Science)	IAPT then HBCSE	10 and below (age rule applies)	Stage 1 NSEJS, Stage 2 INJSO, then selection camp for IJSO **[VERIFY]**	Stage 1 Nov, Stage 2 Jan-Feb **[VERIFY]**
NSEB then INBO (Biology)	IAPT then HBCSE	11-12	Stage 1 NSEB, Stage 2 INBO, then camp for IBO **[VERIFY]**	Stage 1 Nov, Stage 2 Jan-Feb **[VERIFY]**
INSPIRE-MANAK	Dept. of Science and Technology	6-12 (Class 11-12 added; Science stream only for those)	School nominates up to 5 ideas online; district then state then national	Last cycle nominations closed 15 Sep 2025; check inspireawards-dst.gov.in for next cycle
NMMS (scholarship test)	Karnataka: DSERT/KSEAB under Govt. of India scheme	8 (govt, aided and local body schools)	Single written test	Applications Sep-Oct; exam Jan-Feb (last held 1 Feb 2026); 2026-27 date not announced yet
NTSE	NCERT; Karnataka Stage 1 via DSERT	10	Stage 1 state, Stage 2 national	No sign it was discontinued, but sources are thin; 2026-27 notification expected Aug-Oct. Confirm at dsert.kar.nic.in and NCERT **[VERIFY]**

Tip for parents: Olympiads do not decide admission, but they sharpen concepts and give a ladder (school, state, national, international) that fits a Class 6 start.

4. Entrance exams after Class 10 / 12

NEET-UG (National Eligibility cum Entrance Test)	
Conducting body	National Testing Agency (NTA); official site neet.nta.nic.in
Used for	MBBS, BDS, BAMS, BHMS, BUMS, BVSc & AH, and several nursing and allied courses; AIIMS and JIPMER seats also use NEET
Minimum qualification	Class 12 with Physics, Chemistry, Biology/Biotechnology and English. Minimum aggregate in PCB: 50% General, 40% SC/ST/OBC, 45% PwBD (English not counted). The 2026 bulletin wording on this is reported unclear, and state AYUSH rules can differ. **[VERIFY in bulletin]**

Age	Minimum 17 years on 31 Dec of the admission year (31 Dec 2026 for the 2026 cycle). No upper age limit is mentioned in 2026 coverage.
Frequency	Once a year
Tentative month	Registration Feb-Mar; exam normally first Sunday of May. 2026: held 3 May, cancelled 12 May after a paper leak, re-exam scheduled 21 June 2026. Dates can shift. **[VERIFY re-exam completed and 2027 schedule]**
Pattern	180 compulsory MCQs: Physics 45, Chemistry 45, Botany 45, Zoology 45. Total 720 marks. Pen and paper (OMR). Duration reported as 180, 195 or 200 minutes by different sources. **[VERIFY duration]**
Marking	+4 for correct, -1 for wrong, 0 for unattempted
Languages	13 languages including English, Hindi and Kannada; language chosen at application is final

KCET (Karnataka Common Entrance Test)	
Conducting body	Karnataka Examinations Authority (KEA); portals kea.kar.nic.in and cetonline.karnataka.gov.in
Used for	Karnataka seats in B.Pharm, BSc Agriculture, Veterinary-allied, Farm Science, Engineering, Architecture and others. Medical, dental and AYUSH seats in Karnataka use the NEET score with KEA counselling.
Minimum qualification	Class 12/2nd PUC with Physics, Chemistry and Maths/Biology; percentage criteria (commonly 45% General, 40% reserved) and domicile/Kannada rules are in the KEA brochure **[VERIFY]**
Frequency	Once a year
Tentative month	April. 2026: 23-24 April, with a Kannada language test on 22 April for eligible (Horanadu/Gadinadu Kannadiga) candidates
Pattern	Each candidate takes three papers: Physics, Chemistry and either Biology or Maths. 60 MCQs per paper (180 in total), 1 mark each, 80 minutes per paper. Offline OMR, English and Kannada.
Marking	No negative marking
Rank basis	Combination of KCET score and 2nd PUC marks (commonly 50:50) **[VERIFY]**

CUET-UG (Common University Entrance Test)	
Conducting body	NTA, cuet.nta.nic.in **[VERIFY]**
Used for	BSc and other UG programmes at Central Universities and many state/private universities; ICAR agriculture UG admissions also use CUET **[VERIFY]**
Minimum qualification	Class 12 pass; each university sets its own subject requirement
Frequency / month	Once a year; May-June **[VERIFY]**
Pattern	Computer-based. Domain papers include Biology, Chemistry, Physics, Psychology: 50 MCQs per subject, all to be attempted, 60 minutes each. Up to 5 subjects in total including languages and General Test; total 250 marks for 5 subjects. **[VERIFY exact subject combination rules in bulletin]**
Marking	+5 correct, -1 wrong, 0 unattempted

IISER Aptitude Test (IAT)	
Conducting body	IISER consortium, iiseradmission.in **[VERIFY]**
Used for	5-year BS-MS at all seven IISERs; 4-year BS at IISER Bhopal and Tirupati; 4-year BTech at IISER Bhopal
Minimum qualification	Class 12 Science with at least three of Physics, Chemistry, Biology, Maths (Biology not compulsory); minimum 60% aggregate, 55% for SC/ST/PwD
Frequency / month	Once a year; 2026 exam was 7 June (registration closed 13 April)

Pattern	Computer-based, 3 hours, 60 MCQs (15 each: Biology, Chemistry, Maths, Physics), 240 marks
Marking	+4 correct, -1 wrong, 0 unattempted

NEST (National Entrance Screening Test)	
Conducting body	NISER and UM-DAE CEBS
Used for	5-year integrated MSc in Biology and other sciences
Minimum qualification	Class 12 Science with minimum marks (about 60% General) **[VERIFY]**
Frequency / month	Once a year; typically June **[VERIFY]**
Pattern	Computer-based; sections on general, biology, chemistry, physics, maths **[VERIFY details and marking]**

Other routes (short list)

Note: NEET-UG 2026 was disrupted by a paper-leak cancellation, so applicants should follow the NTA and counselling announcements closely each year.

Exam / route	For	Note
AIIMS BSc Nursing entrance	Nursing at AIIMS institutes	Check whether replaced by NEET/CUET in current year **[VERIFY]**
BITSAT	B.Pharm at BITS Pilani	PCB route allowed for B.Pharm **[VERIFY]**
KCET / RGUHS counselling	BSc Nursing and paramedical in Karnataka	KCET (KEA) per an April 2023 government order; NEET reported as not required in Karnataka; 45% PCB General / 40% SC/ST, age 17 by 31 Dec; deemed universities (Manipal, Yenepoya) run their own tests. See Health and Allied document. **[VERIFY on kea.kar.nic.in and rguhs.ac.in]**
Armed Forces Medical (AFMC)	MBBS via armed forces	Uses NEET, then AFMC selection

Postgraduate and beyond (preview)

Exam	Body	For	Tentative timing
NEET-PG	NBEMS	MD/MS after MBBS	Once a year; usually mid-year **[VERIFY]**
NEET-MDS	NBEMS	MDS after BDS	Once a year **[VERIFY]**
GATE (Biotechnology / Life Sciences)	IIT-organised	MTech, PhD, PSU jobs	Feb **[VERIFY]**
IIT JAM (Biological Sciences / Biotechnology)	IIT-organised	MSc at IITs/IISc	Feb **[VERIFY]**
CSIR-UGC NET (Life Sciences)	NTA	JRF, lectureship	Twice a year **[VERIFY]**
GPAT	NTA	M.Pharm	Once a year **[VERIFY]**
CUET-PG	NTA	MSc at central universities	Once a year **[VERIFY]**

5. Scholarships

Scholarship	Provider	Who can apply	Amount	Window
NMMS	Govt. of India via Karnataka DSERT	Class 8 in govt/aided/local body schools; family income up to 3.5 lakh; at least 55% in Class 7 (50% SC/ST) **[VERIFY: single source]**	12,000 per year (1,000 per month), Class 9-12	Apply Sep-Oct; exam Jan-Feb
INSPIRE-MANAK	DST with NIF	Class 6-12, idea nominated by school (max 5 per school)	10,000 one-time	Nominations around Jul-Sep; portal inspireawards-dst.gov.in
INSPIRE-SHE	DST	Top 1% in Class 12 boards, or top 10,000 in JEE/NEET, or KVPY/NTSE/JBNSTS/Olympiad scholars, studying BSc / BS / Int. MSc in natural and basic sciences. No income limit.	80,000 per year (60,000 scholarship + 20,000 summer-project mentorship), renewable	Expected deadline 1 Dec 2026, not officially notified yet; portal online-inspire.gov.in **[VERIFY]**
CSSS (Central Sector Scheme)	Ministry of Education via National Scholarship Portal	Top 20% (above 80th percentile) of successful candidates in own board and stream; family income up to 4.5 lakh	12,000 per year for UG years 1-3; 20,000 per year for years 4-5 / PG	2026-27 portal opened 1 Jun 2026; one listing shows 31 Oct 2026 deadline **[VERIFY]**
Post-matric scholarships	National Scholarship Portal and Karnataka SSP	SC/ST/OBC/minority categories, income limits	Varies	Typically Jul-Nov **[VERIFY]**
Corporate / foundation scholarships	Private foundations	Merit and need based	Varies	Varies by foundation **[VERIFY]**

Note: KVPY was discontinued and its role absorbed by INSPIRE-SHE and NEET/JEE based selection. Confirm before listing it anywhere. **[VERIFY]**

6. Courses and colleges

Course	Duration	Entry	Note
MBBS	4.5 yr + 1 yr internship	NEET-UG	Gateway to doctor career and MD/MS
BDS (Dental)	4 yr + 1 yr internship	NEET-UG	Dentist; MDS later
BAMS / BHMS / BUMS	4.5 yr + 1 yr internship	NEET-UG; AACCC (15%) + state counselling	AYUSH systems; see Health and Allied document
BVSc & AH	About 5 yr	NEET-UG merit (VCI)	Veterinary; 50% PCB (47.5% reserved); see Agriculture document
BSc Nursing	4 yr	KCET + KEA counselling in Karnataka **[VERIFY]**	Hospital and community nursing; see Health and Allied document
B.Pharm / Pharm.D	4 yr / 6 yr	KCET, BITSAT, state CETs	Pharma industry, hospital pharmacy
BPT (Physiotherapy)	4.5 yr	State or NEET **[VERIFY]**	Rehabilitation, sports
BSc Allied Health (MLT, Radiology, OT, Optometry)	3-4 yr	NEET reported mandatory from 2026-27 (NCAHP) **[VERIFY]**	Diagnostics and technician roles
BSc Life Sciences / Biotechnology / Microbiology / Genetics	3-4 yr	CUET-UG or university	Research and industry route
BS-MS (IISER / NISER)	5 yr	IAT, NEST	Research focused
BSc Agriculture / Horticulture / Forestry	4 yr	CUET-UG (ICAR), KCET	Agri-business, research

Course	Duration	Entry	Note
BTech Biotechnology / Biomedical Engineering	4 yr	JEE / KCET	Needs Maths in Class 12 (PCMB only)

Examples of institutions (illustrative, not a ranking): AIIMS and JIPMER, Bangalore Medical College, St John's Medical College, Manipal and Kasturba Medical College, IISc, IISERs, NISER, Christ University, Mount Carmel College, RGUHS-affiliated colleges. **[VERIFY availability and fees]**

7. Career outcomes

Path	Job roles	Higher study
Medicine	Doctor, surgeon, specialist, public-health officer, medical officer in defence	MD/MS, DM/MCh, fellowships
Dental / AYUSH / Veterinary	Dentist, ayurvedic or homeopathic physician, veterinarian	MDS, MD (AYUSH), MVSc
Nursing / Allied health	Nurse, physiotherapist, lab technologist, radiographer, optometrist	MSc, MPT, hospital management
Pharmacy	Pharmacist, drug regulatory, clinical research, pharma industry	M.Pharm, MBA, PhD
Life sciences / Biotech	Researcher, lab scientist, bioinformatician, quality analyst	MSc, PhD, CSIR/ICMR fellowships
Agriculture / Environment	Agri-officer, plant breeder, forest service, food technologist	MSc Agri, PhD, UPSC/State services

8. Related umbrellas this stream connects to

- Design: biomedical and product design (NID DAT and UCEED accept any stream; confirm current eligibility). **[VERIFY]**
- Law: CLAT is open to Science students; medical law and forensic careers combine both.
- Defence: Armed Forces Medical Services and Military Nursing Service through NEET-linked selection; NDA Army wing allows any stream but Air Force/Navy need Maths and Physics. **[VERIFY]**
- Psychology and counselling: CUET-UG with Psychology; PCB students are eligible.
- Public policy and civil services: any graduate can attempt UPSC; Science graduates often choose Medical Science or Zoology as optional subject. **[VERIFY current optional list]**

9. Verification log (version 2, 17 Oct 2026)

What changed after checking by web search. Confidence is based on secondary sources only.

Item	Version 1 said	Now says	Confidence
NEET 2026 schedule	First Sunday of May	3 May exam cancelled 12 May (paper leak); re-exam 21 Jun 2026	High (several news reports); confirm re-exam was completed
NEET pattern	180 Q, 3 hr 20 min	180 compulsory MCQs, 4 x 45, +4/-1 confirmed; duration varies by source (180/195/200 min)	High except duration
NEET languages	Kannada **[VERIFY]**	Confirmed: 13 languages including Kannada	High
NEET age	Upper age **[VERIFY]**	Min 17 by 31 Dec; no upper limit mentioned for 2026	Medium
NEET PCB %	50 / 40 / 45	Same values found; 2026 bulletin wording reported unclear; AYUSH rules vary by state	Medium
KCET pattern	Four papers of 60 Q	Per candidate: Physics, Chemistry + Maths or Biology = 180 Q; 60 each, 80 min each, no negative marking. 2026 dates 23-24 Apr	Medium-High
KCET eligibility and rank basis	Estimated	Still unconfirmed	Low

Item	Version 1 said	Now says	Confidence
CUET-UG	~50 MCQs, 60 min	50 MCQs per domain subject, all to attempt, 60 min, +5/-1, CBT. Exam month not confirmed	Medium
IAT (IISER)	Pattern unknown	60 Q, 15 each, 3 hr, 240 marks, +4/-1; 60% (55% reserved); 7 Jun 2026	Medium-High
NTSE	May be discontinued	No evidence it was discontinued; Karnataka Stage 1 via DSERT	Low-Medium
INSPIRE-MANAK	Class 6-10	Class 6-12 now; 10,000; up to 5 ideas per school	Medium-High
INSPIRE-SHE	~80,000	80,000 = 60,000 + 20,000 mentorship; no income cap; top 10,000 JEE/NEET	Medium
CSSS	~12,000	12,000 (years 1-3), 20,000 (years 4-5); top 20% of board; income up to 4.5 lakh	Medium
NMMS	Nov-Dec test	Applications Sep-Oct, exam Jan-Feb (held 1 Feb 2026); 12,000 per year	Medium; income limit single-source
BSc Nursing (Karnataka)	Unclear	KCET + KEA counselling; NEET not required in Karnataka; 45% / 40%; deemed universities own tests	Medium

Still to verify on official sites

- NEET: PCB percentage rules in the current bulletin, exam duration, and the 2027 schedule.
- KCET: eligibility percentages and rank basis. CUET: exam month and subject-combination rules. NEST: all details.
- Olympiads (SOF, NSEJS, NSEB), NTSE 2026-27, post-matric scholarships, AIIMS nursing, BITSAT, NEET-PG / GATE / JAM / CSIR-NET / GPAT / CUET-PG dates, BVSc and BPT entry route, Design and Defence eligibility.
- Official sources: neet.nta.nic.in, cuet.nta.nic.in, kea.kar.nic.in, cetonline.karnataka.gov.in, iiseradmission.in, scholarships.gov.in, online-inspire.gov.in, inspireawards-dst.gov.in, dsert.kar.nic.in, rguhs.ac.in, nmc.org.in.
- Dates shift every year. Replace "tentative month" with the official notification date before publishing.

ONE-PAGE GIST: Science - PCB

Draft - verify all **[VERIFY]** items in the detailed version before publishing.

Stage	What to do
Class 6-8	Try Olympiads (NSO/IMO); Class 8: NMMS scholarship test
Class 9-10	Take Standard Maths if unsure; Junior Science Olympiad (NSEJS)
After Class 10	Choose PCB (+ Maths or Computer Science optional); PUC in Karnataka
Class 11-12	Boards + NEET-UG; also KCET, CUET-UG, IAT, NEST
After Class 12	MBBS/BDS/AYUSH/Vet via NEET; Nursing, Pharmacy, Allied, Life Sciences, Agriculture, IISER
Postgraduate	NEET-PG, NEET-MDS, GATE, JAM, CSIR-NET, GPAT

Exam	When	Pattern	Marking
NEET-UG	Once a year, May (2026 re-exam: 21 Jun)	180 Q (Phy 45, Chem 45, Bot 45, Zoo 45); 720 marks; about 3 hr 20 min	+4 / -1
KCET	Once a year, April	Phy, Chem + Maths or Bio: 60 Q each, 80 min each	+1 / 0
CUET-UG	Once a year, May-June **[VERIFY]**	Up to 5 subjects, 50 MCQs each, 60 min	+5 / -1
IAT (IISER)	Once a year, June	CBT, 3 hr, 60 Q (15 each: Bio, Chem, Maths, Phy), 240 marks	+4 / -1
NEST	Once a year, June	CBT; multi-section science test	Check notification

Eligibility for NEET: Class 12 PCB + English, minimum 50% aggregate General (40% reserved), age 17+ by 31 Dec. **[VERIFY]**

Scholarships: NMMS (Class 8, 12,000/yr), INSPIRE-MANAK (Class 6-12, 10,000), INSPIRE-SHE (80,000/yr) and CSSS (12,000/yr) after Class 12, post-matric schemes via NSP/SSP. **[VERIFY income limits and deadlines]**

Careers: doctor, dentist, AYUSH physician, vet, nurse, physiotherapist, pharmacist, lab scientist, biotech researcher, agri-officer, public health.

Want a personalised roadmap or counselling? Contact LessonPlan, Yelahanka, Bangalore.